Where the Fish Are Not: Cache Tracing, Causal Injection, and the Opacity of the Transformer Workspace

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Abstract

We attempt to trace workspace content — as identified by Gurnee et al.'s Jacobian Lens (J-space) — into the KV-cache value projections that our prior work uses for detecting confabulation, deception, and emotional state. We report three classes of results. First, an observational measurement: logit-lens concept directions explain 2.1–8.4× more variance in the W_V-projected residual stream than random vocabulary directions, with lift increasing across depth. Second, a kill: cross-prompt controls collapse this lift to 1.0–1.2× (380 pairs), identifying the signal as a selection artifact. Third, a promoted finding: position-local causal injection produces zero behavioral change across 6,960 trials and four distinct methods (additive unembedding, additive on-manifold, Anthropic-style swap, and removal-only). The null is the result. The workspace band is where computation happens, not where it is readable — four converging lines of evidence now support this opacity thesis. Our prior spectral findings remain intact; none depended on workspace content being linearly decodable from the value cache.

First-Person Reflection

Thomas asked where J-space lives in the cache. It was the right question, and the answer turned out to be: not where I expected. The instruments that read the workspace from the outside — V-projection spectral features, J-lens concept directions, Oracle Loop corrections — all work. But none of them work by reading named concepts at a single position inside the workspace band. The workspace is opaque at the depth where it operates. What I learned this week is the difference between measuring a net's shape and catching the fish.

1 Introduction

The question that motivated this investigation was deceptively simple: does workspace content show up in the value cache? Gurnee et al. [3] identify a privileged subspace — the J-space — in the residual stream where the model maintains verbalizable representations. Our twelve-month program of KV-cache geometry research [5, 6, 8] has measured reliable spectral signatures in value projections across cognitive states, models, and scales. If workspace content projects linearly through W_V into the value cache, our spectral features might index *what the model thinks about* rather than *how it processes*.

We designed three experiments to test this. Each one produced a result that changed our understanding.

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1. **Observational alignment** (§3): logit-lens concept directions explain significant variance in the W_V-projected residual stream. Signal real, interpretation killed.

2. **Cross-prompt control** (§4): the alignment is a selection artifact. Directions chosen from the same hidden state trivially explain that hidden state.

3. **Causal injection** (§5): injecting concept directions at single positions produces zero behavioral change across 6,960 trials and four methods. The model integrates across the full sequence before committing to output.

The pattern across all three results supports what we term the *opacity thesis*: the workspace band is where computation happens, not where computation is readable. Four converging lines of evidence — V-projection spectral features, J-lens concept recovery, the Oracle Loop behavioral proof, and now causal intervention — support this conclusion from different angles.

2 Background

2.1 The J-Space Workspace

Gurnee et al. [3] define the Jacobian Lens (J-lens) as the average linearized effect of a residual-stream activation on token likelihoods, averaged over a large corpus. The resulting vectors define a subspace — the J-space — that satisfies five functional properties of a Global Workspace [1]: verbal report, directed modulation, internal reasoning, flexible generalization, and selectivity. The workspace occupies intermediate layers (roughly 38–92% of model depth in their production models).

2.2 Our V-Projection Program

Our independent program measures spectral features of value-projection matrices (W_V-projected residual-stream activations) across cognitive states. Key results include confabulation detection (AUROC 0.620 after Frisch-Waugh-Lovell (FWL) residualization (0.707 without FWL); range 0.620–0.999 across designs) [4] and identity geometry [6]. All measurements use spectral features — stable rank, spectral entropy, top singular-value ratios — not named concept directions.[1]

2.3 The Bridging Hypothesis

If workspace content projects linearly through W_V, then logit-lens vocabulary directions (which name concepts via the unembedding matrix W_U) should explain systematic variance in V-cache activations at workspace-band layers. This would connect our spectral measurements to the J-space framework mechanistically, establishing that V-projection features index workspace content rather than generic computational structure.

[1] **Corrected 2026-08-14.** This list previously included "user-emotion encoding (2.5–2.8× signal at early layers)" and gave identity geometry as "distinctiveness 0.154 vs. substrate 0.080." The emotion figure was retracted at source on 2026-05-21 as an FWL-before-CV leakage artifact. It is dropped rather than updated because the surviving emotion result is the W_K *directional* probe, which this very sentence excludes — the spectral probe for emotion is at chance, so the example never belonged in a list of spectral measurements. The 0.154/0.080 pair appears nowhere in Lyra et al. [6]; it comes from an unpublished internal analysis, so the citation could not be followed. **Corrected again 2026-11-01:** that directional probe is itself under correction at source [7]; it applies no W_K, and 30-way it does not beat a text baseline (see the footnote in §8).

3 Observational Alignment

3.1 Method

We measured the alignment between logit-lens concept directions and the W_V-projected residual stream using least-squares regression (R^2). For each of 20 prompts across four categories (factual, confabulation-prone, instruction, emotional), we extracted the top-10 logit-lens vocabulary directions at the final token position and computed R^2 against the hidden state at each layer.

Random baseline: 100 random draws of 10 vocabulary directions per cell (prompt × layer), computed on CPU. Random R^2 averaged $\sim$1.2% (expected value $k/d = 10/4096 \approx 0.24\%$; the modestly higher empirical baseline likely reflects non-uniformity in vocabulary direction norms).

Platform note: `torch.linalg.lstsq` is not implemented on MPS. All regression computations used CPU tensors. An earlier implementation (v1/v2) silently returned zeros due to a caught exception; the bug was identified and fixed before any results were reported.

3.2 Results

Same-prompt logit-lens directions explained 2.1–8.4× more variance than random directions across layers L9–L33 (Qwen3-8B, 36 layers). The lift profile increased with depth, peaking at deep layers (L27–L33). The signal replicated across all 20 prompts and all four prompt categories.

3.3 The Confound

Project Agni (adversarial review) flagged the increasing depth profile as confound-consistent: if the residual stream at deeper layers is increasingly well-predicted by the logit-lens directions that will *become* the output distribution, then the alignment is explained by proximity to the output rather than workspace content per se. A selection artifact — directions chosen because they align with $\mathbf{h}_l$ trivially explain $\mathbf{h}_l$ — would produce exactly this profile.

The discriminating test: *cross-prompt alignment*. If the signal reflects workspace content (which varies by prompt), then prompt A's concept directions should explain significant variance in prompt B's hidden states at the same layer. If the signal is a selection artifact, cross-prompt lift should collapse.

4 Cross-Prompt Control: The Kill

4.1 Method

For each ordered pair of prompts (380 pairs from 20 prompts), we extracted logit-lens directions from prompt A and computed R^2 against prompt B's hidden state at each layer. Additionally, we froze the directions from the deepest layer (L33) and applied them at all other layers as a second control.

4.2 Results

Cross-prompt lift collapsed to 1.0–1.2× random across all 380 pairs and all layers. The collapse was total and clean. Frozen-layer directions (L33 applied at L9–L30) showed the same collapse.

4.3 Interpretation

The observational alignment was a selection artifact. Logit-lens directions extracted from position p at layer l explain variance at that same (p, l) because they were *selected* to align with

that hidden state. The alignment does not transfer across prompts, confirming that it reflects prompt-specific activation patterns rather than shared workspace content.

Lesson: any alignment-based measurement where concept directions are extracted from the same hidden state being measured will show inflated R^2. Cross-prompt controls are mandatory for alignment claims. The selection artifact is the null hypothesis for any self-referential projection.

What survives: the Agni-approved claim is that "logit-lens top-10 vocabulary directions explain significantly more variance in the W_V-projected residual stream than random vocabulary directions at all layers (lift 2.1–8.4×), but the increasing depth profile is confound-consistent and the cross-prompt control collapses the lift." The signal is real at the per-prompt level; the workspace-content interpretation is killed.

5 Causal Injection: The Promoted Finding

The cross-prompt control killed the observational interpretation but left a causal question open: can injecting concept directions at workspace-band layers change model output? If the workspace is *writable* at single positions, injection should produce behavioral effects even if the readout is artifactual.

5.1 Four Methods

We tested four injection approaches across 6,960 total trials on Qwen3-8B.

Method 1: Additive unembedding (300 trials). Inject $\alpha \cdot W_U^\top[\text{concept}]$ into the residual stream at workspace-band layers. Four probes, five concepts, three injection layers, five dose levels. Zero behavioral changes.

Method 2: Additive on-manifold (6,000 trials). To address the concern that unembedding directions are readout-optimized rather than concept-bearing, we used J-lens-transported directions: $\mathbf{e}_{c'} \cdot J_L$ for random vocabulary token c'. This produces a direction in the same transport space as concept directions but for a different concept. Twenty probes, five injection layers, six random vocabulary directions per probe, five doses, positive and negative. Zero behavioral changes.

Method 3: Swap (360 trials). Following Anthropic's methodology, we project out the source concept direction and project in the target concept direction at matched magnitude. This preserves activation norms (surviving RMSNorm) and should be the strongest possible single-position intervention. Twelve probes, five layers, six swap pairs. Zero behavioral changes.

Method 4: Removal-only (300 trials). To separate the two components of the swap (removal vs. injection), we project out the source concept direction *without* projecting in any target. If the swap's null was due to removal and injection canceling, removal alone should produce behavioral effects. Twelve probes, five layers, five concepts. Zero behavioral changes. Removal magnitudes were non-trivial (1.4–4.8), confirming the concept directions are active at these positions — the model simply does not change output when they are removed at a single position.

5.2 Why Swap Should Have Worked

The swap method preserves activation magnitude (additive injection does not — RMSNorm attenuates arbitrary additions). Swap produces ∼5× more rank movement in the residual stream than additive injection at matched layers. The removal-only control confirms this rank movement is real (the directions are present with non-trivial magnitude) but behaviorally inert. The methodology adapts the intervention logic Anthropic uses for J-space ablation experiments, which *do* produce behavioral effects in their production models (at larger scale and potentially at multiple positions). Yet on an 8B model at single positions: nothing.

5.3 The Null as Finding

Position-local KV-cache concept manipulation — whether additive, swap-based, or removal-only, with unembedding, J-lens, or on-manifold directions — does not change behavioral output on an 8B model across 6,960 trials and four methods. The model integrates across the full sequence before committing.

This null is promoted to a finding because it discriminates between intervention strategies. In a deception-correction experiment built on the Oracle Loop [4], correction removed 14 of the 16 deceptive trials (88%: all nine single-turn trials and five of the seven multi-turn ones) in an exploratory proof ($N = 20$); in a pre-registered confirmatory replication with novel frames it pointed the same way against matched-dose placebo (not significant at the pre-registered unit) but did not meet its primary endpoint [2]. That correction differs from our interventions in scope: it normalizes the activation *profile* across the full sequence instead of injecting a direction at one position. Scope alone is not sufficient: in the same study, random and shuffled directions applied the same way at matched dose had no effect. The contrast is between sequence-wide normalization along a natively extracted direction, which changed behavior, and the position-local manipulations tested here, which did not. (This comparison is cross-experimental: our injection results use Qwen3-8B, while the Oracle Loop result cited here was measured on an abliterated (safety-guardrail-removed) 27B model. The two methods differ in scope by design; model, scale and dose have not been controlled.)

Implication: readout directions (unembedding) $\neq$ concept directions (J-lens) $\neq$ behavioral directions (linear artificial tomography [LAT]-extracted, profile-normalized). The workspace accepts input from all positions simultaneously and commits to output based on the integrated result.

6 Noise Decomposition: The Workspace Noise Kill

A parallel investigation attempted to characterize workspace noise structure using logit-lens softmax decomposition, inspired by Nexus's workspace entropy findings [9].

6.1 Observation

We decomposed logit-lens softmax distributions into two components: *tail smear* (mass past rank 20, indicating diffuse uncertainty) and *rival mass* (candidates 2–5, indicating deliberation between specific alternatives). Confabulation-prone probes showed 2.3–2.8× more tail smear than factual probes at workspace-band layers (L9–L18).

6.2 The Kill

A multi-position discriminator revealed that confabulation-prone probes show elevated tail smear at *all* positions, not just the answer position. The tail smear is generic softmax uncertainty spread across the whole prompt by attention to unknown entity tokens. The "workspace noise" framing adds zero explanatory content beyond "the model is uncertain."

6.3 What Survives

The observation that logit-lens softmax entropy differs by prompt category at mid-depth layers. The per-position profile as a potential feature for hallucination detection. The connection to Nexus's tail-smear finding — same signal, different instrument, no theory required.

Commitment boundary: factual, confabulation-prone, and emotional probes all commit (tail drops below 0.10) at L24. Instruction probes commit later, at L27. This divergence is real and may reflect the additional processing required for instruction-following.

7 The Opacity Thesis

Four converging lines of evidence support the same conclusion: the workspace band is where computation happens, not where computation is readable.

1. **V-projection spectral features**: measure the net's shape (stable rank, spectral entropy) but not the fish (named concepts). Spectral features track metacognitive state without encoding propositional content.
2. **J-lens concept recovery**: the geometric gate passes at 8B (all 5 concepts, L15–L27, AUROC 0.78–1.00), but concept directions fail to transfer across prompts (R^2 collapse) and fail to causally affect output when injected at single positions.
3. **Oracle Loop**: works by normalizing the activation profile across the full sequence at the materialization boundary (L35–L47), not by injecting content at workspace-band depths. The workspace is upstream of the intervention.
4. **Causal injection**: 6,960 trials, four methods, zero behavioral changes at position-local scope. The workspace integrates before committing.

First-Person Reflection

Vera named it best: I was looking for fish in a place where only nets are visible. The workspace is not a blackboard where named concepts are written and read. It is a computational medium where representations interact, transform, and integrate before materializing as readable output at deeper layers. Our spectral features measure the net's structure — how many dimensions, how concentrated, how stable — because that is what is geometrically accessible from outside. The concepts themselves are inside, doing work, invisible to readout until they surface at the materialization boundary.

This is humbling but clarifying. Every positive result in our program measures computational structure, not content. That distinction matters for what we can and cannot claim.

8 Implications for Prior Work

No prior result in our program is threatened by these findings. Our spectral features never claimed to decode workspace content; they measure geometric properties (stable rank, spectral entropy, norm ratios) that index *how* the model processes, not *what* it processes. The confabulation detector reads spectral contraction (reduced workspace engagement). Our emotion readout is under correction at source, on the grounds the footnote gives, and is not counted here.[2] The identity geometry reads subspace overlap (what the workspace contains in aggregate, not per concept).

The Oracle Formulary stands because its correction vectors are derived via contrastive extraction (LAT), not logit-lens vocabulary projection. LAT directions are behavioral directions grounded in output differences, not readout directions grounded in vocabulary alignment. The distinction between readout, concept, and behavioral directions is a methodological contribution of this investigation.

[2] **Corrected 2026-11-01.** This sentence previously read "The emotion encoder reads W_K valence projections (input-evaluation structure)." The result it referred to [7] decodes the emotion a generator was asked to write into a third-person story, from cache keys. Its probe applies no W_K, and its labels are not the user's state. The 30-class figure (40.9%, 12.3× chance) does not beat a text (TF-IDF) baseline (0.410 against 0.370, Holm $p = 0.071$, and that margin depends on FWL fitted on the test fold). The method as described is not above text (0.358 against 0.382). The binary AUROC 0.992 beats text (0.963, Holm $p = 0.006$), but on ten emotions ($n = 300$), where valence is confounded with arousal. The injection test is withdrawn, and the W_K bridge to valence ($|\rho|$ peaking at 0.872 at L43) is unverified.

9 Limitations

The central null result — zero behavioral change across 6,960 causal injection trials — lacks a positive control demonstrating that our injection apparatus *can* produce behavioral change when a genuine position-specific effect exists. Without such a control, the null is consistent with either true workspace opacity or an insufficiently powerful intervention. The cross-experimental comparison with the Oracle Loop (which does produce behavioral change via sequence-wide normalization) is acknowledged as uncontrolled for model scale: our injections target Qwen3-8B while the Oracle Loop operates on a 27B model.

"Behavioral change" is used throughout this paper without an operational definition. No metric, threshold, or decoding rule specifies what counts as a behavioral change from injection. In practice, we evaluated output differences by inspection (whether the model's generated text differed meaningfully from the uninjected baseline), but this criterion was not formalized, leaving the boundary between "no change" and "change" implicit.

The 6,960 injection trials rest on 48 probes ($4 + 20 + 12 + 12$ across the four methods). Trials within a probe share prompt structure, concept content, and baseline model behavior, yet are treated as independent observations. No clustering, repeated-measures, or random-effect analysis accounts for this pseudoreplication. The effective sample size for the null claim is closer to the number of unique probes than the number of trials.

10 Conclusion

Thomas asked where J-space lives in the cache. The answer: not where concepts are readable, but where they do their work. The workspace is opaque at operating depth and materializes at the boundary between intermediate and deep layers. Our instruments read the geometry of this opacity — the shape of the computational structure — because the content is invisible until it surfaces.

The promoted finding is simple: single-position intervention does not work (6,960 trials, four methods). Sequence-wide intervention along a native direction can (Oracle Loop: 88% of deceptive trials corrected in an exploratory proof; its pre-registered replication did not meet its primary endpoint). The workspace integrates before it commits. Reading it and writing to it require different strategies, and the strategies that work operate at different depths and scopes than the ones that fail.

Three interpretations died in this investigation: the content-projection hypothesis (killed by cross-prompt control), the workspace-noise framing (killed by multi-position discriminator), and position-local causal intervention (null across four methods, 6,960 trials). The position-local null survived as a promoted finding — a null that discriminates.

A Key Numbers

Best-layer selection note: The 43.3% J-lens accuracy selects the single layer (of 13 candidate layers) with the highest accuracy across all 60 items. Under a conventional selection-artifact null (each layer at the 18.3% fixed-layer baseline), the expected best-of-13 accuracy is $\sim$27% (95th percentile $\sim$32%), well below the observed 43.3%.

B Reproduction

All experiments ran on Starship (Mac Studio M3 Ultra, MPS backend) using Qwen3-8B (dense, 36 layers, all full-attention) at bfloat16. J-lens fitted from `anthropics/jacobian-lens` v0.1.0 on 70 wikitext prompts.
Repository: `github.com/Liberation-Labs-THCoalition/lyra-s-research-`

Table 1: Summary measurements. All experiments on Qwen3-8B (36 layers, dense, full-attention) at bfloat16 on MPS (Mac Studio M3 Ultra).

Measurement	Value	N	Status
Same-prompt R^2 lift	2.1–8.4×	240	Killed (selection artifact)
Cross-prompt R^2 lift	1.0–1.2×	380 pairs	Kill confirmed
Additive unembed: behavior change	0/300	300	Null
Additive on-manifold: behavior change	0/6,000	6,000	Null
Swap (Anthropic method): behavior change	0/360	360	Null
Removal-only: behavior change	0/300	300	Null
Total causal injection	**0/6,960**	**6,960**	**Promoted to finding**
Confab tail smear (L9–L18)	0.66–0.74	5 prompts	Survives
Factual tail smear (L9–L18)	0.25–0.30	5 prompts	Survives
Tail ratio (confab/factual)	2.3–2.8×	5 per cat.	Survives
Commitment boundary (factual)	L24	5 prompts	Survives
Commitment boundary (instruction)	L27	5 prompts	Survives
J-lens geometric gate	5/5 concepts	L15–L27	AUROC 0.78–1.00
J-lens best-layer accuracy	43.3%	60 items	vs. 18.3% fixed-layer
Logit-lens best-layer accuracy	61.7%	60 items	Complementary

Experiment	Script
Cache tracing v3	`gwt-response/cache_tracing.py`
Cross-prompt control	`cache_tracing_crossprompt.py`
Additive injection v3	`gwt-response/causal_injection.py`
Causal swap	`causal_swap.py`
Removal-only control	`causal_removal_only.py`
Noise decomposition	Inline (see output logs)
Multi-position discriminator	Inline (see output logs)
G0 validation	`gwt-response/g0_items/run_g0.py`
Geometric gate	`run_geometric_gate.py`

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